ATOM COUNT CARD

Mole Concept and Conversion

PRODUCT IDENTITY



Product Name:

CAFF Caffeinated Energy Bite, Dark Chocolate

Compound Name:

Caffeine

Formula:

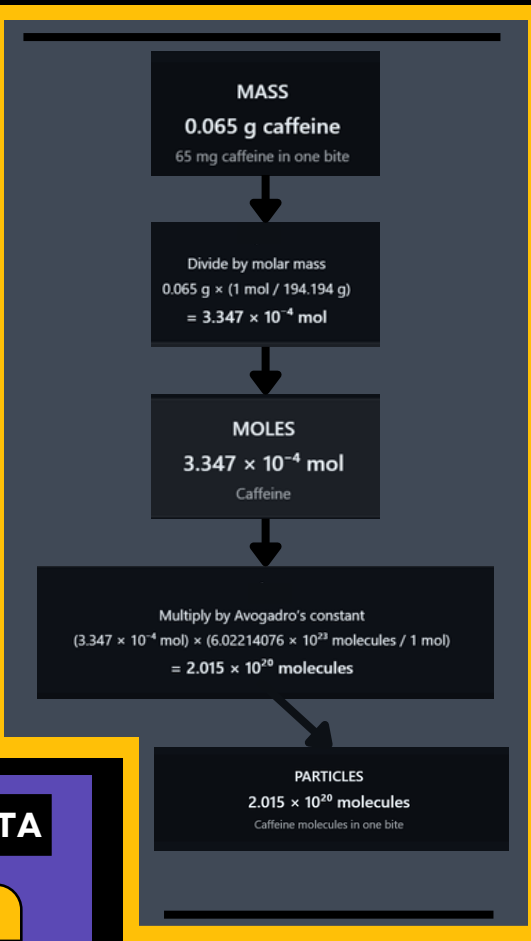
$C_8H_{10}N_4O_2$

Why the Compound is in the Product:

Caffeine is included as a stimulant.

Product source:

https://eatcaff.com/products/energy-bites



MOLE MAP

ESQUILADOR, JOSE MIGUEL T.

BASE QUANTITATIVE DATA

Quantity	Value/Units	Source	Precision
Caffeine per bite	65 mg/bite	Manufacture's product info	2 sig. fig.
Pack size used	20 bites/pack	Manufacturer's listed pack option	Exact
Scale- up scenario	20 bites	Assumption: 20 students, one bite each	Exact
Formula of Caffeine	$C_8H_{10}N_4O_2$	PubChem	Exact Subscript
Atomic masses	C -12.011, H- 1.008, N- 14.007, O- 15.999	CIAAW	Not limiting
Avogadro's constant	$6.02214076 \times 10^{23}$ mol^{-1}	IUPAC (2019)	Exact

MOLAR MASS BUILD-UP

Elements	Atoms	Atomic mass (g/mol)	Contribution (g/mol)	Percent by Mass
C	8	12.011	96.008	49.480%
H	10	1.008	10.080	5.191%
N	4	14.007	56.028	28.852%
O	2	15.999	31.998	16.477%
Total	24	43.025	194.194	100%

MOLE CONVERSION CALCULATIONS

Molar mass of $C_8H_{10}N_4O_2$:

8(12.011) + 10(1.008) + 4(14.007) + 2(15.999) = 194.194 g/mol

Label bridge: caffeine per package

20 bites × (65 mg caffeine / 1 bite) = 1,300 mg caffeine

1,300 mg × (1 g / 1,000 mg) = 1.30 g caffeine

Mass → moles (one bite)

65 mg × (1 g / 1,000 mg) × (1 mol / 194.194 g)
= 3.347×10^{-4} mol caffeine

Moles → molecules

3.347×10^{-4} mol × ($6.02214076 \times 10^{23}$ molecules / 1 mol)
= $2.015 \times 10^{20} \approx 2.0 \times 10^{20}$ caffeine molecules

Atoms of one element inside the formula

2.015×10^{20} molecules × (4 N atoms / 1 molecule)
= $8.06 \times 10^{20} = 8.1 \times 10^{20}$ nitrogen atoms

Percent composition + two-route check (Nitrogen)

$$\%N = \frac{4(14.007)}{194.194} \times 100 = 28.85\%\ N$$

Route 1: Using percent composition

$$65\ mg\ caffeine \times \frac{28.85mg\ N}{100mg\ caffeine}$$

= 18.75 mg N ≈ 19 mg N

Route 2: Using moles and atomic mass

0.065 g caffeine × (1 mol caffeine / 194.194 g caffeine) ×
(4 mol N atoms / 1 mol caffeine) × (14.007 g N / 1 mol N atoms)

= 0.01875 g N × (1,000 mg / 1 g)
= 18.75 mg N ≈ 19 mg N

Both agrees

REVERSE: FROM % COMPOSITION TO THE FORMULA

Assume a 100 g sample of caffeine **($C_8H_{10}N_4O_2$).**

Element	Mass%	Mass (g)	Moles (mol)	Atomic Mass (g/mol)	Simplest Ratio
C	49.48%	49.48g	4.118	12.011	4
H	5.19%	5.19g	5.149	1.008	5
N	28.85%	28.85g	2.0160	14.007	2
O	16.48%	16.48g	1.030	15.999	1

Calculate moles

Moles = mass ÷ atomic mass

C: 49.48 ÷ 12.011 = 4.118 mol
H: 5.19 ÷ 1.008 = 5.149 mol
N: 28.85 ÷ 14.007 = 2.060 mol
O: 16.48 ÷ 15.999 = 1.030 mol

Find the simplest ratio

Divide each mole value by the smallest value, 1.030.

C_4	H_5	N_2	O
4	5	2	1

Empirical formula: $C_4H_5N_2O$
Molecular formula: $C_8H_{10}N_4O_2$

Reverse: how much CAFF energy bite product contains one mole of caffeine?

$$1.00\ mol \times \frac{194.194}{1\ mol} \times \frac{1\ bite}{0.065g\ caffeine} \times \frac{1\ package}{20\ bites} = 149.38\ packages$$

In the % composition of nitrogen atoms, 28.85% was used as the equivalent of 100% to determine the number of nitrogen atoms in caffeine ($C_8H_{10}N_4O_2$).

SCALE-UP: FROM ONE PACKAGE TO THE WHOLE CAMPUS

1. ONE PACKAGE
1.3g of caffeine
6.70×10^{-3} mol
4.04×10^{21} molecules

2. ONE CLASSROOM, ONE DAY
Assume 20 students, 1 package/student
0.134 mol
8.07×10^{22} molecules

3. CAMPUS, ONE YEAR
Assume 1,000 students, 100 packages/year
670 mol
4.04×10^{26} molecules

4. REVERSE
One mole of caffeine is in
149.38 packages
of CAFF energy bites.

SIGNIFICANT-FIGURE CHECK

Calculator: 2.0155... × 10^{20} molecules
Reported: 2.02 × 10^{20} molecules

- Limits: 65 mg (2 s.f.), Exact N_a, Exact subscripts.
- 65 mg limits the final answer to 2 s.f.
- Therefore, the final reported value is rounded to 2 s.f.

ONE-NUMBER TAKEAWAY

"One mole of caffeine is equivalent to eating exactly 149.38 packages of CAFF energy bites."

REFERENCE

CAFF, Inc. (n.d.). Caffeinated energy bite. https://eatcaff.com/products/energy-bites

National Center for Biotechnology Information. (n.d.). PubChem compound summary for CID 2519, caffeine. PubChem. https://pubchem.ncbi.nlm.nih.gov/compound/2519

International Union of Pure and Applied Chemistry. (2019). Atomic weights of the elements 2019. https://www.qmul.ac.uk/sbcs/iupac/AtWt/

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